Ethan R. Partida

CONTACT INFORMATION

Department of Mathematics
Brown University
Providence, RI 02903

Email: ethan_partida[at]brown[dot]edu
Web: ethanpartida.github.io

INTERESTS

Algebraic Combinatorics, Algebraic Geometry, Convex Geometry, Tropical Geometry, Combinatorial Optimization

EDUCATION

Brown University, Providence, Rhode Island USA
Ph.D. student in Mathematics, Expected Spring 2027
Advisors: Melody Chan and Caroline Klivans

University of Minnesota, Minneapolis, Minnesota USA
Bachelor's degree in Mathematics, May 2022
Bachelor's degree in Computer Science, May 2022

PAPERS

The Chow Polynomial of the Noncrossing Partition Lattice (with J. Chun, N. Hayashida, and Z. Wang)
arxiv.2608.03806, Submitted for Journal Publication

Hodge theory for combinatorial projective bundles (with M. Larson)
arxiv.2604.21925, Submitted for Journal Publication

Algebraic Cobordism rings of Wonderful Varieties and Matroids (with R. Gandhi)
arxiv.2606.12645 Submitted for Journal Publication
Séminaire Lotharingien de Combinatoire, FPSAC 2026 Extended Abstract

Graded Ehrhart Theory of Unimodular Zonotopes (with C. Crowley)
arxiv.2603.07873 Submitted for Journal Publication
Séminaire Lotharingien de Combinatoire, FPSAC 2026 Extended Abstract

Line Shellings of Geometric Lattices (with S. Backman, G. Dorpalen-Barry, A. Nathanson, and N. Prime)
arxiv.2601.05119 Preprint (Revisions in progress, Fall 2026)
Séminaire Lotharingien de Combinatoire, FPSAC 2026 Extended Abstract

Shifted and threshold matroids. Discrete Applied Mathematics, Volume 378, 2026

On the topology of conormal complexes and posets of matroids (with A. Nathanson). SIAM Journal on Discrete Mathematics, Volume 40, Issue 1, 2026

Playing Cribbage with Reinforcement Learning and Minimax. University of Minnesota Conservancy

Computing Rational Powers of Monomial Ideals. (with P. Dongre, B. Drabkin, J. Lim, E. Roy, D. Ruff, A. Seceleanu, and T. Tang) *Journal of Symbolic Computation*, Volume 116, 2023

AWARDS

2025-2027 Simons Dissertation Fellowship

2025,2026 ×2 GAANN Graduate Fellowship Recipient

2024 NSF Graduate Research Fellowship Honorable Mention

2021 MAA Outstanding Student Paper Presentation

2021 JMM Honorable Mention Poster

2020 Ronald E. McNair Scholar

2020 Prof. Hans H. Dalaker Mathematics Undergraduate Scholarship

2020 William H. Burgum College of Science and Engineering Scholarship

2020 Texas State Rapid Fire Research Winner

2019-2020 ×2 National Hispanic Scholarship Fund Recipient

2019 MN Space Grant Recipient

2019 3M Diversity Scholarship

Advising and Mentoring

Brown University, Summer UTRA Advisor of Thea Miller-Priddy, Haoyang Xu, Patrick Whiteford, and Zakhar Pysarevskyi (with Melody Chan), Summer 2026

Brown University, Undergraduate Senior Thesis Advisor of Natsuka Hayashida (with Bena Tshishiku), Fall 2025-Spring 2026

Brown University, Summer UTRA Advisor of Joseph Chun, Natsuka Hayashida, and Zhixing Wang (with Melody Chan), Summer 2025

Brown University, Undergraduate Senior Thesis Advisor of Ameer Muhialdeen (with Melody Chan), Fall 2024-Spring 2025

×2 Brown University, Graduate Student Mentor (of a First Year Graduate Student), 2023-2025

×6 Brown University, DRP Mentor, Spring 2023-Current

University of Minnesota, Mathematics Project at Minnesota, Peer Mentor, Fall 2021

Professional Activities

Referee for the journal Algebraic Combinatorics, 2026

AMS Eastern Sectional Meeting, Organizer of the Special Session on Group Actions and Combinatorics in Algebraic Geometry and Commutative Algebra, Spring 2026

Brown DRP, DRP Organizer, Fall 2024-Current

Brown Horizons Seminar, Organizer, 2023-2026

Brown Horizons, Reading Group Organizer, Spring 2023

OURFA^{2}M^2, Experiences Panel Panelist, Fall 2022

University of Minnesota, Undergraduate Mathematics Research Seminar, Organizer, Fall 2021

University of Minnesota, SMART Learning Commons, Peer Tutor, Summer 2020-2022

University of Minnesota, North Star STEM Alliance, Member, Fall 2019-2022

Selected Presentations

2026 University of Washington Combinatorics and Geometry Seminar

2026 UC Berkeley Combinatorics Seminar

2026 UC Davis Combinatorics Seminar

2026 20 Minute Talk, Special Session on Algebraic and Geometric Aspects of Matroid Theory, AMS Fall Central Sectional Meeting

2026 Dartmouth Combinatorics Seminar

2026 University of Toronto Combinatorics Seminar

2026 North Carolina State University Algebra and Combinatorics Seminar

2026 Lightning Talk, Algebra and Geometry of Matroids Workshop, Casa Matemática Oaxaca

2026 MIT/Harvard Richard P. Stanley Seminar in Combinatorics

2026 Poster Session, Formal Power Series and Algebraic Combinatorics (FPSAC)

2026 University of Oregon Combinatorics Seminar

2026 Graduate Online Combinatorics Colloquium

2026 UCSD Combinatorics Seminar

2026 University of Minnesota Combinatorics Seminar

2026 20 Minute Talk, Commutative Algebra Meets Algebraic Combinatorics (CAAC)

2024 20 Minute Talk, VIII Encuentro Colombiano De Combinatoria

2024 Brown University Combinatorics Seminar

2024 20 Minute Talk, Santander Workshop on Geometric and Algebraic Combinatorics

2022 PME Poster Session, Joint Mathematics Meetings

2021 Lightning Talk, Texas State Graduate Expo

2021 Student Talk, Young Mathematicians Conference,

2021 Student Paper Session, MAA Mathfest

2021 Student Presentation, Brown University's SUMS

2021 MAA Poster Session, Joint Mathematics Meetings

Teaching

Brown University Teaching Assistant
MATH 90 Calculus I, Fall 2023
MATH 90 Calculus I, Spring 2024
MATH 190 Calculus II, Fall 2025
Brown University Instructor
MATH 520 Linear Algebra with Team Based Learning, Spring 2027